

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

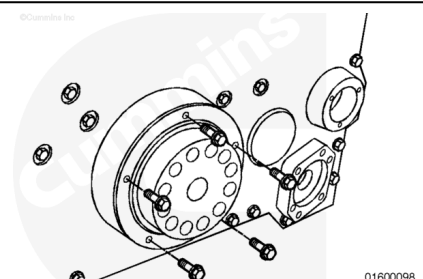
This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



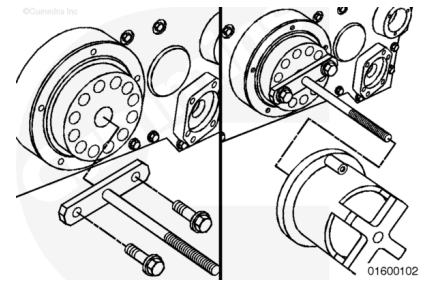
- Remove the fan and fan spacer(s). Refer to Procedure 008-040 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-040-tr.html)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html)
- Remove the vibration damper(s) and fan drive pulley. Refer to Procedure 001-052 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-052-tr.html)
- Remove the crankshaft adapter. Refer to Procedure 001-017 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-017-tr.html)

Remove

Remove the five front crankshaft seal mounting capscrews.



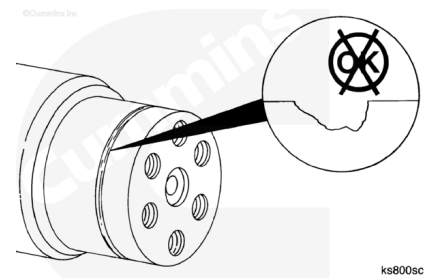
Remove and discard the front crankshaft seal from the front gear cover with the Front and Rear Crankshaft Seal Installation and Removal Tool Kit, Part Number 3163349.



Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the crankshaft at the seal contact area. Use a clean cloth and solvent.

Install

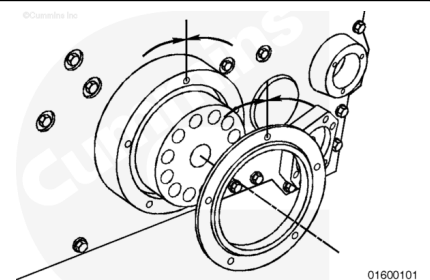
Note : Do not separate the crankshaft seal from the carrier during installation. This could cause damage to the seal faces and scrolls resulting in premature failure of the seal and oil leakage.

Note : The use of a guide pin will make seal alignment more accurate.

Note : Do not remove the seal from its packaging unless you are able to start and finish the installation in one action. Leaving the seal exposed out of its packaging can result in dirt ingress into the seal and result in premature failure of the seal and oil leakage.

Apply a light coating of clean engine oil to lubricate the rear rubber diameter of the seal.

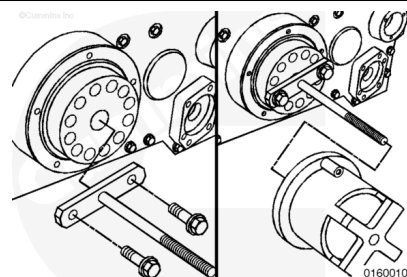
Install the crankshaft seal carrier onto the crankshaft dry. Do **not** lubricate it.



Use hand pressure to push the rear crankshaft seal onto the crankshaft as far as possible.

⚠ CAUTION ⚠

Do not use a hammer on the tool. Damage to the threads will result. The tool can bind to the crankshaft because of the narrow tolerances.

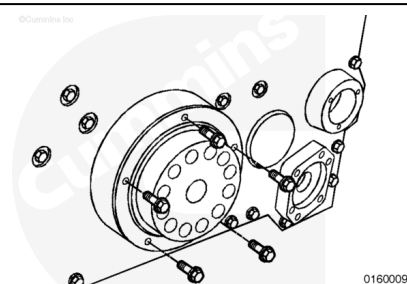


Use the Front and Rear Crankshaft Seal Installation and Removal Tool Kit, Part Number 3163349, to install the front crankshaft seal.

Install the five front crankshaft seal mounting capscrews.

Tighten the capscrews alternately and evenly.

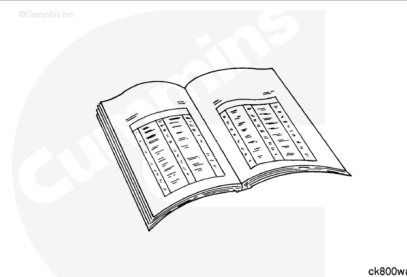
Torque Value: 10 n·m [89 in-lb]



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To reduce the possibility of serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the crankshaft adapter. Refer to Procedure 001-017 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-017-tr.html>)
- Install the vibration damper(s) and fan drive pulley. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-052-tr.html>)
- Install the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-002-tr.html>)

- Install the fan spacer(s) and fan. Refer to Procedure 008-040 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-040-tr.html)
- Operate the engine and check for leaks.

Last Modified: 23-Feb-2016
